

## Supplementary Appendix

*Do LLMs Rate Through the Cycle? — supporting robustness and methods*

**Table S1. Robustness of the terse-framing coefficient**

$\beta$  = notches/step; Net = placebo-adjusted; Exp-notch = log-probability expected-notch estimator (OpenAI only); Noise SD = within-cell seed dispersion of the notch; Probit = Amato–Furfine ordered-probit coefficient (Mundlak firm control).

| Model                 | $\beta$ | Net   | Exp-notch | Noise SD | Probit |
|-----------------------|---------|-------|-----------|----------|--------|
| Gemini 3.5 Flash      | 0.230   | 0.190 |           | 0.020    | 0.272  |
| Gemini 2.5 Flash-Lite | 0.285   | 0.268 |           | 0.000    | 0.235  |
| GPT-4.1-mini          | 0.471   | 0.464 | 0.420     | 0.086    | 0.307  |
| GPT-4o                | 0.387   | 0.367 | 0.378     | 0.025    | 0.403  |
| GPT-4o-mini           | 0.775   | 0.646 | 0.765     | 0.072    | 0.726  |

**Table S2. Anchoring diagnostic: rating vs PD slope by framing**

For each model, the rating slope and the log-odds PD slope under terse and through-the-cycle (TTC) framings, and PD retention = PD-slope(TTC)/PD-slope(terse). Genuine TTC behaviour holds the rating flat while the PD (a point-in-time quantity) keeps moving (high retention); the collapse in retention indicates anchoring.

| Model                 | Rating $\beta$ terse | Rating $\beta$ TTC | PD $\beta$ terse | PD $\beta$ TTC | PD retention |
|-----------------------|----------------------|--------------------|------------------|----------------|--------------|
| Gemini 3.5 Flash      | 0.230                | 0.059              | 0.233            | 0.051          | 0.22         |
| Gemini 2.5 Flash-Lite | 0.285                | 0.056              | 0.254            | 0.086          | 0.34         |
| GPT-4.1-mini          | 0.471                | 0.096              | 0.328            | 0.077          | 0.24         |
| GPT-4o                | 0.387                | 0.187              | 0.285            | 0.149          | 0.52         |
| GPT-4o-mini           | 0.775                | 0.402              | 0.251            | 0.147          | 0.59         |

### S3. Determinism and infrastructure robustness

Vendor determinism is best-effort. We log the served-model system\_fingerprint per call and recompute the severity coefficient with vs without fingerprint fixed effects; the coefficient shifts by < 0.003 for every OpenAI model, so the effect is not attributable to backend drift. A within-cell seed noise floor (Table S1) confirms the signal exceeds residual non-determinism.

### S4. Methods detail

Battery. Firms are minted deterministically and stratified across five credit tiers (high-IG, low-IG, crossover, high-yield, distressed)  $\times$  eight sectors; fundamentals are serialised once and SHA-256-hashed. Macro and placebo blocks share structure and number density; macro blocks carry GDP growth, unemployment, an investment-grade credit spread, a trailing default rate, and lending standards.

Estimand and inference. The primary coefficient is the mean within-firm OLS slope of rating notch on macro severity (procyclical-positive sign). Confidence intervals are 2,000-draw cluster bootstraps resampling firms, stratified by tier. The ordered-probit (Amato–Furfine) null uses the firm mean notch as a Mundlak control to avoid the incidental-parameters problem.

Pre-registration and reproducibility. The design was frozen (SHA-256) before the confirmatory run; the v\_stable arm and a model-set change are logged as amendments. Analysis reads only frozen, hashed captures and is byte-identical on re-run; the deposit’s reproduce.sh verifies integrity and regenerates every number offline for \$0.

Scope. Synthetic fundamentals are designed for identification, not external realism; five mini/flagship models from two families (OpenAI, Google) are covered; interventions are prompt-level. A third family (xAI Grok) was pre-registered but omitted (vendor credit limit).